# Bengali Speech Sentiment Analysis using Machine Learning Models: A Comparative Study

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***Abstract*—Though scholars find speech sentiment analysis based on audio data to be a very intriguing study topic, not enough work has been done for the fifth most spoken language in the world, Bangla. The purpose of this study is to close this research gap. SUBESCO, BanslaSER, and KBES combined dataset were utilized in this study's evaluation of all the models. Evaluations have been done on CNN models, ML models, and sequence models like LSTM and BI-LSTM. Mel-frequency spectrum was exploited by CNN models, while machine learning and sequence models were applied to numerical features. DenseNet201 achieved the finest accuracy of all the models, at 94%. Finally, utilizing both hard and soft voting, all of the models based on numerical features and spectrogram features were ensembled, yielding accuracy rates of 94.99% for numerical features and 95% for spectrogram features.**

***Keywords*—SSA, SUBESCO, BanslaSER, KBES, CNN, ML, Sequence model, Spectrogram, MFCC, Ensemble**

## I. Introduction

Opinion mining, or sentiment analysis, is the computational process of determining and extracting sentiments, opinions, or emotional tones expressed in various types of data, including textual, audio, or audio-visual data. Sentiment analysis is an essential function in a variety of systems which includes product review analysis, consumer feedback analysis, education engagement analysis, danger detection systems, etc. Throughout recent years, researchers have widely explored speech analysis of sentiment using text-based information. As social media has grown in popularity and user base, it has become an asset of textual corpus. However, sentiment analysis based on audio data is to be explored thoroughly. Because human speech can convey a wide range of emotions through the fusion of acoustic features, linguistic elements, and prosody, it is crucial to explore sentiment analysis based on audio data.

This trend continues the low resource languages such as Bangla. Most of the research done with Bengali Speech sentiment analysis works on textual data. Considering the number of Bengali speakers this is a crucial research gap that requires investigation and this research works in reducing the corresponding flaw by evaluating performances of ML, DNN, and RNN on features extracted from a Bengali speech dataset. This dataset has been created by merging three datasets KBES, BanglaSER, and KBES. Different features such as the Mel-Frequecy Spectrogram, Chroma Shift, Zero Crossing Rate, Mel Frequency Cepstral Coefficients, etc. have been extracted from the datasets. More emphasis has been put on Mel-Frequncy Spectrograms as since its introduction by S.Davis and P.Mermelstein [1], it has become a staple in sentiment analysis.

## II. Related Works

Analyzing speech sentiment is a comparatively new and demanding research sector as people use various types of voice and audio in day-to-day scenarios. Though there are many studies on finding and optimizing the efficiency of SSA in various languages, most of the studies are done in English. Speech Sentiment Analysis in the Bangla language carries a vital significance as it is the fifth most spoken language in the world [2]. Even though Bengali is one of the most spoken languages, there has been comparatively less research done on this language than English.

Among them, [3] thoroughly examined different models for SSA using Bengali datasets like SUBESCO and BanglaSER. There, the datasets were experimented with CNN architecture like VGG-19, Inception v4, and ResNet-50 as well as RNN like LSTM and Bi-LSTM models. They also implemented ML models like Random Forest, AdaBoost and K-Nearest Neighbour. The results of the study showed that RF and KNN models had the highest accuracy rates (90%) followed by LSTM and Bi-LSTM models (80%), while AdaBoost and CNN models performed worse (60%). However, in the custom noisy dataset, the accuracy drastically decreased, indicating how different the models' performance was in different scenarios.

For SSA, there are many papers where SUBESCO and RAVDESS datasets have been used for research. Among them, [4] concentrated on choosing significant features to increase accuracy. According to the study, enhanced feature selection can produce results with accuracy that are comparable to deep learning models but at a lower computational cost. In their paper, different ML models, including RF, SVM and XGBoost, were implemented. Their model's optimized feature set achieved an accuracy of 82.9% for SUBESCO using SVM, demonstrating the efficacy of their method in

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the recognition of emotions. On the other hand, [5] concentrated on Bengali audio speech emotion recognition where capturing both local and sequential speech characteristics, the research created a novel model that merges a deep convolutional neural network (DCNN) with a bidirectional long-short term memory (BLSTM) network, enhanced with a time-distributed flatten layer. Known as DCTFB, this model outperformed other CNN-based models with high accuracy rates of 86.9% on SUBESCO and 82.7% on RAVDESS. A thorough cross-lingual analysis using transfer learning was also included in the study, demonstrating the model's wider linguistic applicability. But most importantly, [6] outperformed these previous works by designing a hybrid model and achieving 98.97% accuracy in SUBESCO and 98.30% accuracy in RAVDESS dataset. For the hybrid model, they integrated one-dimensional CNN with an LSTM architecture into a fully connected network. This paper strongly highlights with evidence why their hybrid model is effective in emotion detection and classification in voice inputs.

While SUBESCO and RAVDESS datasets were popular, [7] worked with a self-collected audio recording dataset, the Abeg dataset. In the research paper, they portrayed a model for identifying emotions in Bengali speech with a focus on neutrality, happiness, and rage. The study compared several ML models, for example, Logistic Regression and SVM, using MFCC as well as LPC features. Significant accuracy was attained by the model; on the Abeg dataset, Logistic Regression performed the best, with an accuracy of 92%. The study also demonstrated how well MFCC and LPC features combined can predict emotions in Bengali speech. In the research, the RAVDESS dataset was utilized to benchmark the effectiveness of the system. When RAVDESS and Abeg both datasets were combined, the XGBoost classifier achieved 86% accuracy. Another research [8] focused on identifying Bengali speech's four emotional states: happy, sad, angry, and neutral. The study trained a k-Nearest Neighbor classifier using feature vectors that included pitch and Mel-frequency Cepstral Coefficient (MFCC). There were four hundred emotive sentences from fifty people in the dataset. With individual emotion detection accuracy of 80% for happy, 75% for sad, 85% for angry, and 75% for neutral emotions, the method yielded an average accuracy of 87.50%.

Meanwhile all the other Banlga SER datasets did not include intensity measure, [9] created a dataset named KBES that includes intensity measure which is comparatively more realistic to other datasets. The study developed a cascaded model which includes CNN with TDF, LSTM, Bi-LSTM and dense layers. The proposed model was trained using 3D transformed speech data which was created by blending Mel-Frequency Cepstral Coefficient, Short-Time Fourier Transform (STFT), and Chroma STFT signal transformation techniques. This was used to both classify the audio sentiment and also intensity. The model reached an accuracy of 88.30% and 71.67% for RAVDESS and KBES respectively.

Upon reviewing the existing researches there remains some noteable limitations to work on. Bangla being a low-resource language, few datasets are made available for classifying speech sentiments. Although some datasets have previously been combined to achieve a larger trainable dataset to work on, some of the most recent datasets still remained un-employed in the combination process. Different approaches have been employed with different datasets for the existing research works. However, no particular benchmark can be found for the approaches to be generalized for the best approach for a generalized dataset. Most existing dataset fail to represent real world scenarios, which vary in audio quality and include speech with different dialects, local Bengali language.

## III. Methodology

### A. Dataset

Three pre-existing audio datasets named KBES [10], SUBESCO [11], and BanglaSER [12] were collected. The overall proposed diagram is represented in Fig. 2.

KBES dataset consists of 900 audio files of following categories: Angry (Low), Disgust (Low), Happy (Low), Sad (Low), Neutral, Angry (High), Disgust (High), Happy (High) and Sad (High). Among these nine types, each of them are of 3 seconds length. Additionally, SUBESCO is a purely audio emotive speech corpus including seven thousand sentence-level Bangla statements. The classes of the audio files are happiness, surprise, anger, fear, neutral, sadness and disgust. In this audio dataset, each file is 3 seconds long. On the other hand, BanglaSER which contains 1467 Bangla speech-audio recordings, have five human sentimental classes, called happy, surprise, neutral, sad, and angry. Each of the audio file of this dataset also is of 3 seconds of length.

With a total of seven distinct classes, SUBESCO has the highest number of sentiment data. Since all five classes of BanglaSER align with those of SUBESCO, the datasets were merged according to their respective classes. KBES has five different sentiments with high-low divisions for each sentiment except Neutral. These high-low divisions were first unified and each class was then combined with the matching classes from SUBESCO. Classwise frequency distribution of different datasets along with our merged dataset is outlined in Table I.

The merged dataset contains 9367 audio files. The highest amount of data is to be found in Happy, Angry and Sad as theese three are common classes of the three datasets. The lowest amount of data is to be found in Fear class as it only exists in SUBESCO dataset.

### B. Feature Extraction

Various feature extraction algorithms were implemented to recover features from the augmented data after it had been enhanced with noise, pitch shifting, and time stretching. Numerous numerical features such as MFCCs, ZCR, Chroma Shift, etc. along with the Log Mel-Spectrograms of the audio

TABLE I. Frequency Distribution of Different Datasets

| Sentiment | No of samples | | | Merged Dataset |
|---|---|---|---|---|
| | ***SUBESCO*** | ***BanglaSER*** | ***KBES*** | |
| Happy | 1000 | 306 | 200 | 1506 |
| Surprise | 1000 | 306 | - | 1306 |
| Angry | 1000 | 306 | 200 | 1506 |
| Sad | 1000 | 306 | 200 | 1506 |
| Fear | 1000 | - | - | 1000 |
| Disgust | 1000 | - | 200 | 1200 |
| Neutral | 1000 | 243 | 100 | 1343 |
| **Total** | **7000** | **1467** | **900** | **9367** |


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data were extracted from the dataset using Python's Librosa library.

From the audio dataset, 40 unique mfcc values were recovered using Librosa library [13]. A signal's sign shifting rate from positive to negative and vice-versa, creating intermediate frequencies is known as the zero-crossing rate or ZCR. We used Librosa to extract the zero crossing for every frame. A different extraction method called CS was applied for assessing the tonal variations of compressed sound data. Also, the chromagram feature from librosa is utilized to help identifying audio's chords and determining harmonic similarity. Root Mean Square (RMS) was also extracted from the audio files using the same library. Finally, a CSV file was created based on the extracted numerical features of all the merged augmented and raw audio data.

Librosa library was also used in generating Log Mel-Spectrograms from the merged audio data, both augmented and raw. Mel-Spectrograms are visual portrayals of frequency spectrums that vary across time mapped onto the Mel scale. To generate the logarithmic Mel-Spectrograms, a logarithmic scale is applied on the frequency axis and a Mel scale is applied to the amplitude axis. The log-mel spectrograms were generated using 128 Mel-bins, Hann window of size 2048 samples, and Hop-length of 512 samples as represented in Fig. 1. The generated Log Mel-Spectrograms of both raw and augmented audio data were stored together in a directory with each spectrogram being in .png format.

### C. Data Augmentation

All three datasets namely used in this research consist of audio samples recorded in a controlled environment free from significant noise and disruptions. This does not represent a real-world environment where a plethora of noise may be present. Thus it was decided to augment the audio files of the dataset by injecting noise, shifting pitch, and stretching time.

The merged dataset was augmented using Librosa. It is a Python package for music and audio analysis developed by Mcfee et.al[13]. Three factors were present in the augmentation of our data.

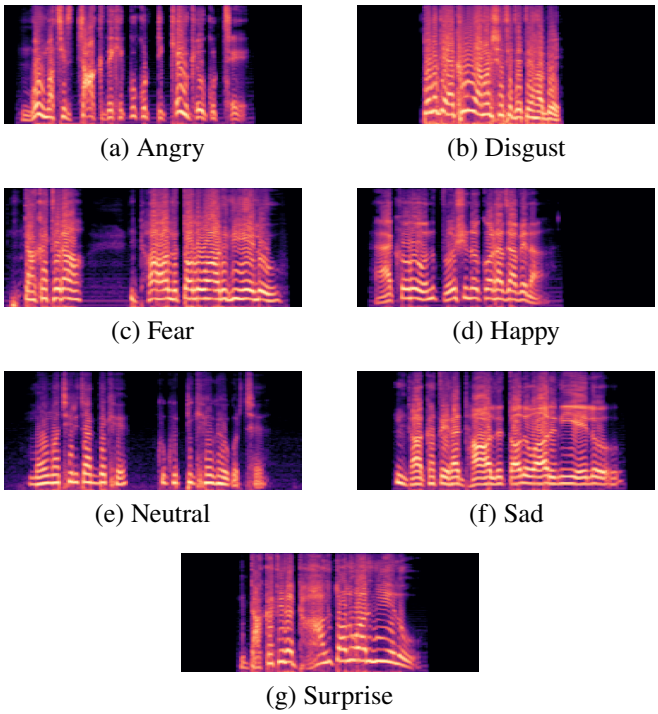
**Pitch Shifting:** The frequencies of the audio files were shifted by a factor of approximately 0.7. A random variance of -0.1 to 0,1 was added to the pitch factor for to randomize each augmentation.

**Time Stretching:** The Time stretching factor was set to 0.8. Also, a random variance of -0,1 to 0,1 was added for randomization of each augmentation.

**Noise Injection:** Noise factor was set to 0 to 0.5% of the maximum amplitude (loudest sound) of each audio data. Gaussian distribution was used for the noise injection and numpy's random function was used to randomize the factor.

### D. Resampling and Normalization

Since the dataset, which is being used, is a combination of three distinct datasets, certain classes of data are absent in some datasets. For example, the fear class was absent in BanglSER and KBES, resulting in a noticeable dataset imbalance. The numerical features of the dataset have been balanced using Synthetic Minority Oversampling Technique (SMOTE). SMOTE balances the dataset by synthesizing new data for the minor class. Additionally, standardization technique has been used to normalize the dataset to ensure uniform scaling among the features.



(a) Angry (b) Disgust (c) Fear (d) Happy (e) Neutral (f) Sad (g) Surprise

Fig. 1. Log Mel-Spectrogram for Different Speech Sentiment Data

### E. Sentiment Analysis Based On Numerical Data using ML Classifiers

Different machine learning models were trained on numerical features extracted from the audio data for the sentiment analysis task. One such algorithm is Random Forest which is a machine-learning ensemble model that employs bootstrap aggregation and decision trees as foundational learners. Decision trees are simple models that aid in decision-making and serve as foundations for more sophisticated ensemble techniques. KNN is a powerful supervised algorithm used for regression and classification, but it can be computationally intensive. Gradient Boost is an iterative method for developing weak learners into strong prediction models, whereas Ada Boost enhances weak learners' performance by assigning weights to training instances. In terms of sentiment analysis from audio speech, these ML models effectively classify emotions by analyzing audio features like MFCCs, ZCR, and so on. These models capture complex emotional patterns in speech, allowing for precise sentiment prediction using subtle sound signals. All the parameters used in these models have been fine-tuned for the highest metrics.

### F. Sentiment Analysis Based On Log-Mel Spectrogram using Convolutional Neural Networks

In another approach, the effectiveness of several popular Convolutional Neural Network (CNN) models trained on Log-Mel Spectrograms for speech sentiment classification was evaluated. ResNet architectures, such as ResNet-50, ResNet-101, and ResNet-152, are suited for capturing subtle features in a variety of domains because they process complex data efficiently by utilizing deep layers with shortcut connections. The


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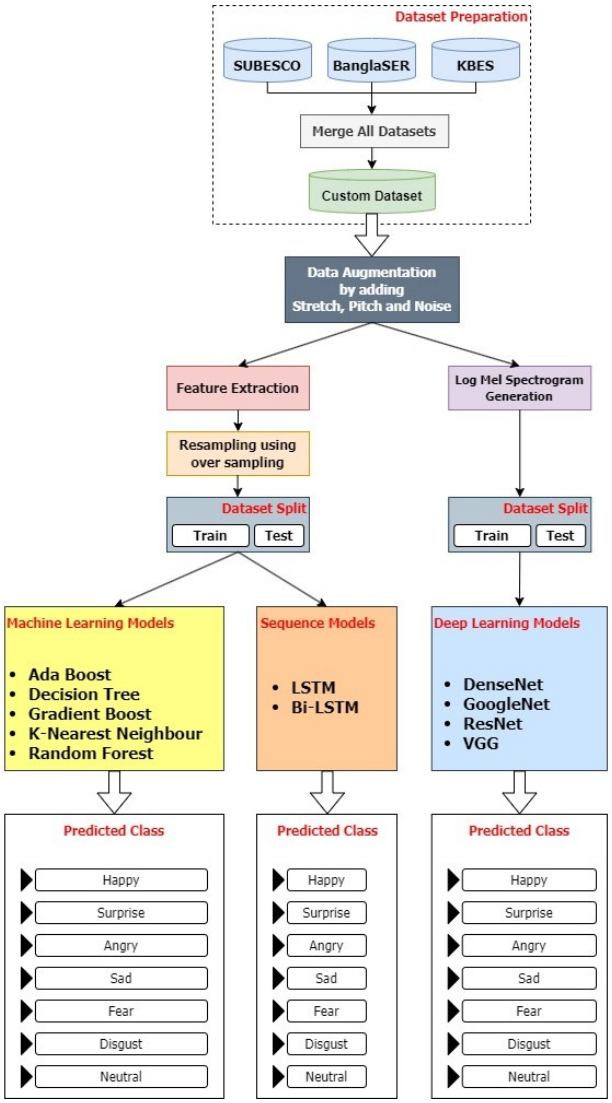



Fig. 2. Flow Diagram of The Proposed Approach

VGG-16 and VGG-19 models emphasize simplicity and depth in feature extraction by using deep stacks of convolutional layers with small filters. With the introduction of GoogleNet's inception modules, extensive data analysis at various scales is made possible by parallel processing using filters of various sizes. With its distinctive densely connected architecture, DenseNet-201 improves network-wide information flow and feature reuse. These architectures make use of effective data processing techniques and deep learning principles to provide strong frameworks for handling challenging tasks. In sentiment analysis with audio speech, these models use log mel-spectrograms to analyze complex sound patterns and identify emotional signals in speech. This method allows for a more detailed understanding of sentiments in spoken speech, which improves the accuracy of emotion detection. In this research, many pre-trained CNN models like ResNet, DenseNet, VGG, and GoogleNet have been used. To better fine-tune the models for this specific task, the top layer of each of the models has been replaced with a custom layer containing five custom-defined residual blocks and a linear layer as the output layer. For VGG models adaptive average pooling was used and then flattened before passing the output to the residual blocks.

### *G. Sentiment Analysis Based On Numerical Data using Sequence Models*

In yet another approach, sentiment analysis based on numerical data was explored in employing different sequence models. Sequence models such as LSTM, a recurrent neural network, enhance the learning and retention of long-term dependencies in sequential data. It is appropriate for tasks such as natural language processing and time series prediction. Bi-LSTM, an extension of LSTM, incorporates data from both past and future contexts. It uses two LSTM layers to process the sequence forward and backward, allowing the model to capture dependencies in both directions. Bi-LSTM is used in natural language processing and speech recognition to make accurate predictions and classifications. In sentiment analysis from audio speech, LSTM, and Bi-LSTM models efficiently detect emotional variations using extracted numerical features such as MFCCs, CS, and so on. These models analyze speech's temporal dynamics, allowing precise emotion detection based on the speech. In this research, a three-layer LSTM network was employed. The layers comprised 100, 50, and 20 cells sequentially.

### *H. Preparing Dataset for Training*

From the combined dataset 10% data have been separated as test dataset. The remaining dataset have been augmented to create the augmented train set which will be used in training. Table II shows the class wise distribution of audio data for each of the subsets of data.

## IV. Result Analysis

### *A. Result Analysis of ML and CNN Models*

To explore the challenging task of sentiment analysis different types of features ranging from numerical features such as ZCR, RMS to Spectrogram features such as Log-Mel Frequency Spectrogram have been extracted. Based on these features, DNNs, ML and sequence models have been applied.

The results obtained from ML and DL models for different features have been compiled in Table III. Popular machine learning models such as Random Forest Classifier, AdaBoost Classifier, KNN, Decision Tree, and Gradient Boost Classifier have been trained on the various features extracted from the different datasets. Moreover, sequence models such as LSTM and Bi-LSTM have been evaluated with the same audio features. Several widely recognized metrics, including Precision, Accuracy and F1-Score have been assessed for all

TABLE II. Class Distribution of Training and Testing Dataset

| Sentiment | Merged Dataset | Test Set | Train Dataset | Train Dataset After Augmentation |
|---|---|---|---|---|
| Happy | 1506 | 151 | 1355 | 2710 |
| Surprise | 1306 | 131 | 1175 | 2350 |
| Angry | 1506 | 151 | 1355 | 2710 |
| Sad | 1506 | 151 | 1355 | 2710 |
| Fear | 1000 | 100 | 900 | 1800 |
| Disgust | 1200 | 120 | 1080 | 2160 |
| Neutral | 1343 | 110 | 1233 | 2466 |


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TABLE III. EVALUATION SCORE OF DIFFERENT ML AND SEQUENTIAL MODELS FOR DIFFERENT FEATURES

| Features | Metrics | Machine Learning Models | | | | | Sequence Models | |
|---|---|---|---|---|---|---|---|---|
| | | *K-NN* | *Random Forest* | *Gradient Boost* | Ada Boost | *Decision Tree* | *BI-LSTM* | *LSTM* |
| MFCCs | Accuracy | 55.27% | 80.83% | 73.69% | 24.49% | 73.16% | 77.95% | 79.76% |
| | Precision | 55.07% | 80.77% | 73.46% | 10.24% | 73.90% | 79.11% | 80.406% |
| | Recall | 55.27% | 80.83% | 73.69% | 24.49% | 73.16% | 77.95% | 79.765% |
| | F1-Score | 55.05% | 80.59% | 73.39% | 11.90% | 73.17% | 75.80% | 77.55% |
| Zero Crossing Rate | Accuracy | 14.32% | 14.44% | 16.71% | 13.41% | 16.08% | 17.57% | 17.25% |
| | Precision | 14.58% | 14.69% | 14.33% | 5.8% | 16.52% | 12.34% | 13.65% |
| | Recall | 14.37% | 14.48% | 14.48% | 13.41% | 16.08% | 17.67% | 17.25% |
| | F1-Score | 14.45% | 14.56% | 14.46% | 5.9% | 16.20% | 10.72% | 10.46% |
| Root Mean Square | Accuracy | 18.21% | 17.57% | 20.44% | 17.41% | 17.145% | 20.23% | 21.19% |
| | Precision | 18.68% | 18.29% | 22.44% | 10.02% | 17.78% | 17.41% | 18.21% |
| | Recall | 18.21% | 17.57% | 20.44% | 17.14% | 17.14% | 20.23% | 21.19% |
| | F1-Score | 18.26% | 17.69% | 17.01% | 7.8% | 16.96% | 12.22% | 13.72% |
| Chroma Shift | Accuracy | 14.90% | 14.90% | 20.76% | 15.01% | 15.76% | 20.55% | 21.61% |
| | Precision | 15.33% | 15.27% | 18.40% | 4.9% | 16.9% | 12.58% | 10.377% |
| | Recall | 14.90% | 14.90% | 20.76% | 15.01% | 15.76% | 20.55% | 21.61% |
| | F1-Score | 15.04% | 15.02% | 15.80% | 6.9% | 15.98% | 20.553% | 13.79% |
| Combined Features | Accuracy | 71.13% | 87% | **90.84**% | 71.139% | 72.94% | 80.29% | 79.44% |
| | Precision | 57.29% | 88.25% | **91.16**% | 57.86% | 73.49% | 81.11% | 79.59% |
| | Recall | 71.13% | 87% | **90.84**% | 71.13% | 72.5% | 80.29% | 79.44% |
| | F1-Score | 61.78% | 87.36% | **90.78**% | 62.04% | 72.86% | 78.01% | 77.12% |
| ZCR+RMS+CS | Accuracy | 28.22% | 33.65% | 32.58% | 17.25% | 35.67% | 31.73% | 33.01% |
| | Precision | 28.34% | 33.66% | 32.97% | 6.08% | 36.95% | 30.67% | 27.7% |
| | Recall | 28.22% | 33.65% | 32.58% | 17.25% | 35.67% | 31.73% | 33.01% |
| | F1-Score | 28.23% | 33.52% | 30.91% | 7.7% | 35.89% | 28.39% | 29.47% |

models. Additionally, performance on different audio features on the different ML and DL models has been assessed. Also, a combination of Zero Crossing Rate, Root Mean Square and Chroma Shift have been evaluated.

CNN models have been trained on Log-Mel Spectrogram extracted from the audio dataset. Popular CNN models such as ResNet50, ResNet101, ResNet152, VGG16, VGG19, DenseNet201 and GoogleNet were chosen for the experiment. While ResNet Family showed almost similar results, comparatively GoogleNet performed worst with an accuracy of 86%. On the other hand, the best-performing model here is DenseNet201 having an accuracy of 94%. Table IV depicts the performance score of CNN models for classifying the speech sentiment.

Combination of all the features yielded the best results having an accuracy of 90.84% with Gradient Boosting and 80.295% on Bi-LSTM. Also as individual feature, MFCCs performed the best across the ML and sequence models, with an accuracy of 80.33% on Random Forest and 77.95% on Bi-LSTM. In addition, it is clearly observed from the evaluation result that individual features such as Chroma Shift, Root Mean Square and Zero Crossing Rate do not perform well on any of the models.

TABLE IV. EVALUATION SCORE OF CNN MODELS USING LOG-MEL SPECTROGRAM

| CNN Models | | | | |
|---|---|---|---|---|
| *Model* | *Precision* | *Recall* | *F1-Score* | *Accuracy* |
| Resnet50 | 90% | 90% | 89% | 90% |
| Resnet101 | 90% | 90% | 90% | 90% |
| Resnet152 | 92% | 92% | 92% | 92% |
| VGG16 | 90% | 90% | 90% | 90% |
| VGG19 | 91% | 91% | 91% | 91% |
| **DenseNet201** | **94**% | **94**% | **94**% | **94%** |
| GoogleNet | 86% | 86% | 86% | 86% |

### *B. Result Analysis of Ensemble approach*

For further analysis, different models were combined using ensemble methods to evaluate their performance on Bengali speech sentiment analysis. Table V illustrates the performance score of the ensemble technique applied to different models. At first, all the CNN models have been ensembled using the voting technique. Both variants of voting ensemble hard voting and soft voting have been evaluated. Both gave the same result with an accuracy of 95%. Similarly, ML models and sequence models have been ensembled together with the same set of numerical features. Two experiments have been done ensembling on all the features and ensembling based on MFCC. Ensembling on all the features provide the best accuracy of 94.99%.

TABLE V. EVALUATION SCORE FOR THE ENSEMBLE TECHNIQUE ON DIFFERENT MODELS

| Model | Feature | | Hard Voting | Soft Voting |
|---|---|---|---|---|
| ML + Sequence Models | All Features | Accuracy | 94.99% | 91.58% |
| | | Precision | **95.52**% | 91.84% |
| | | Recall | 94.99% | 91.58% |
| | | F1-Score | 94.93% | 91.36% |
| | MFCC | Accuracy | 83.91% | 82.42% |
| | | Precision | 84.28% | 82.95% |
| | | Recall | 83.91% | 82.42% |
| | | F1-Score | 83.71% | 81.69% |
| CNN Models | Log-Mel Spectrogram | Accuracy | **95.00**% | **95.00**% |
| | | Precision | 95.00% | 95.00% |
| | | Recall | **95.00**% | **95.00**% |
| | | F1-Score | **95.00**% | **95.00**% |


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## V. Conclusion and Future Works

After the analysis, it can be concluded that among CNN, ML, and Sequence Models, for Speech Sentiment Analysis, Convolutional Neural Networks outperform the others. Ensemble approaches were also explored which increased the accuracy on both the numerical and spectrogram-based approaches. Both the approaches converged to 95% accuracy. This provides evidence to the fact that ensemble-based approach is necessary to increase accuracy in this topic. The research highly contributes to the field of artificial intelligence in engineering and technology, addressing an important gap in speech sentiment analysis for one of the world's most spoken yet low-resource languages. This work provides a benchmark to determine effective approaches for speech sentiment analysis by comparing different multiple ML and deep learning models as well as showcases the usefulness of hybrid approaches by utilizing voting ensemble methods. It provides valuable insights for researchers working on speech sentiment analysis, especially for low-resource languages. There has been almost no research conducted by combining all three available Bangla SER datasets up until now. In this research, all three existing datasets have been combined to form a generalized enlarged custom dataset on which all the comparative models have been trained and evaluated. In this research, no custom dataset including real-world noise was involved, which is not depictive of the real-world scenario. There is much more room for work in this regard as more CNN models including self-developed ones might be evaluated to see the results. Robust models using noisy real-world data with varied speaking styles may be developed based on the ground works laid by this research. The effectiveness of using other appropriate features in deploying the compared approaches may be explored for a better understanding of which method best suits such classification. In short, this work can work as a foothold in starting a plethora of research.

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